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SHAPING THE FUTURE WORKFORCE: STUDENTS' PERCEPTIONS ON AI AND HUMAN-CENTRIC TECHNOLOGIES IN INDUSTRY 5.0

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Abstract

Research purpose. This paper aims to analyse how university students perceive the role of Artificial Intelligence and human-centric technologies in shaping the future workforce within the context of Industry 5.0. By focusing on students as future professionals, the study explores their expectations, ethical concerns, and readiness for AI-driven transformation in a developing country context.

Design / Methodology / Approach. A total of 344 university students from the disciplines of Information Technology, Computer Engineering, and Business Management participated in the study through an online questionnaire. The survey assessed students' knowledge of AI technologies, their perceptions of AI's role in workforce transformation, and their ethical concerns related to AI integration. Data were collected from Bachelor's and Master's students enrolled at public and private higher education institutions in Albania. The study complied with institutional research ethics regulations. As it involved an anonymous, voluntary survey without the collection of sensitive or personally identifiable data, formal ethics committee approval was not required. Spearman's Rank Correlation and descriptive statistics were used to analyse the relationships between AI knowledge, workforce perceptions, and ethical concerns.

Findings. The findings reveal that students generally recognize AI's transformative potential in reshaping industries and workforce dynamics, particularly among Information Technology and Computer Engineering students. However, they express scepticism regarding AI's ability to enhance human-centric traits such as creativity and emotional intelligence. A statistically significant positive correlation was found between AI knowledge and perceptions of industrial transformation and skill augmentation, indicating that higher digital literacy is associated with greater optimism about AI's role. Conversely, students who voiced stronger ethical concerns—especially regarding algorithmic bias, privacy, and job displacement—also placed greater emphasis on the need for transparency and explainability in AI systems. These patterns reflect the relevance of both the Technology Readiness and Acceptance Model (TRAM) and Trust in Automation Theory, underscoring the importance of integrating ethical reasoning with technical training to prepare students for responsible engagement with AI in the context of Industry 5.0.

Originality / Value / Practical implications. While student perceptions of AI have been widely examined in global contexts, this study offers a distinct contribution by focusing on Albania—a developing country underrepresented in the AI and Industry 5.0 literature. Its originality lies in the integration of two behavioural frameworks—the Technology Readiness and Acceptance Model (TRAM) and Trust in Automation Theory—to explain students' attitudes toward AI within a human-centric industrial paradigm. The research draws on insights from students in Information Technology, Computer Engineering, and Business Management, integrating both ethical and educational dimensions. These findings provide practical implications for designing inclusive, ethically grounded, and future-ready strategies in education and workforce development.

Keywords: AI integration; industry 5.0; workforce evolution; human-centric technologies; ethical AI

JEL codes: L86; O33; I23

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Introduction

The rise of Artificial Intelligence (AI) marks a pivotal moment in the evolution of industrial and economic systems. This transformation is particularly evident within the emerging framework of Industry 5.0, which shifts the focus from the automation-driven model of Industry 4.0 to one that emphasizes collaboration between humans and intelligent systems (Ghobakhloo et al., 2024; Pizon & Gola, 2023). Industry 5.0 builds upon the foundations of earlier industrial revolutions, incorporating cutting-edge technologies to drive innovation, improve productivity, and unlock new sources of economic value. By enabling smarter automation, data-driven decision-making, and the creation of personalized solutions, Industry 5.0 promises to revolutionize sectors such as manufacturing, healthcare, finance, and logistics.

The integration of AI within Industry 5.0 holds significant potential for shaping the future workforce (Ivanov, 2022). AI technologies can optimize resource allocation, enhance operational efficiency, and stimulate innovation, enabling businesses to create more sustainable and efficient solutions (Chowdhury et al., 2023; Gamberini & Pluchino, 2024; Ghobakhloo et al., 2024). As AI and human collaboration reshape business processes, Industry 5.0 also fosters new roles, skills, and ways of working that reflect a more human-centric approach (Kolade & Owoseni, 2022). While these advancements bring significant benefits, they also present challenges, particularly concerning job displacement, the ethical consequences of AI-driven automation, and the increasing demand for transparent and explainable AI systems (Giroux et al., 2022; Liu et al., 2024; Maddikunta et al., 2022). The large-scale adoption of AI also raises issues related to data privacy, algorithmic fairness, and the appropriate level of human oversight in AI decision-making (Binns, 2018).

Understanding how AI and human-centric technologies will shape the future workforce in Industry 5.0 is crucial for businesses, policymakers, and educators (Lu, et al., 2022). These stakeholders are tasked with navigating this transformative era, making it essential to gain insights into students' perspectives. Education plays a vital role in preparing the future workforce for these changes. Education 5.0, which integrates AI and other advanced technologies into the learning process, ensures that students develop not only technical skills but also the creativity, critical thinking, and adaptability needed in an AI-driven world (Tavares et al., 2022). As Industry 5.0 emphasizes human-machine collaboration, the education system must adapt to nurture these essential skills, preparing students for careers where they work alongside AI rather than being displaced by it. The aim of this study is to examine how university students perceive the role of AI and human-centric technologies in shaping the future workforce within the context of Industry 5.0. By focusing on students as future professionals, the study explores their expectations, ethical concerns, and readiness for AI-driven transformation in a developing country context. Notably, this research contributes to the literature by aligning student perspectives with the emerging principles of Industry 5.0 and by offering region-specific insights into workforce readiness and technology adoption. To explore these topics, this research will address the following questions:

- How do students perceive AI's potential to influence the future workforce and drive transformation across industries in the context of Industry 5.0?
- What are students’ views on the ethical challenges associated with AI, such as job displacement, privacy risks, and algorithmic bias, within Industry 5.0?

Based on the literature and objectives of this study, the following hypotheses are proposed:

- H1: Students with greater knowledge of AI technologies will report more positive perceptions of AI’s potential to transform industries and enhance human skills within the context of Industry 5.0 (α=0.05)
- H2: Students with stronger concerns about the ethical implications of AI are more likely to view transparency and explainability in AI systems as critical to the success of Industry 5.0 ($\alpha = 0.05$).

Drawing on survey data and existing literature, this paper will offer actionable recommendations for businesses, policymakers, and educators to facilitate the ethical, transparent, and inclusive integration of AI into Industry 5.0, ensuring a balanced future workforce.

Literature review

Industry 5.0 introduces a human-centred industrial model, leveraging advanced technologies like AI, automation, and smart systems to enhance human capabilities, with a strong focus on worker well-being, productivity, and sustainability (Adel, 2022). The European Union's vision for Industry 5.0 emphasizes resilience, sustainability, and human-centric values, aiming to balance technological advancements with ethical and social considerations (European Commission, n.d.). It envisions work environments where humans collaborate with technology, rather than being replaced by it, ensuring job satisfaction, safety, and autonomy. However, challenges like job displacement, ethical dilemmas, and technological inequalities remain, requiring solutions that promote fairness, transparency, and inclusivity in technology development (Adel, 2022; Giroux et al., 2022; Lu, et al., 2022). To successfully implement Industry 5.0, fostering human-AI collaboration and addressing socio-technical tensions is crucial (Pacheco & Iwaszczenko, 2024). AI and robotics drive this vision, improving workplace efficiency and worker engagement (Leng et al., 2022). However, their success requires continuous reskilling, education, and policy frameworks to ensure equitable transitions in the workforce (Longo et al., 2020).

AI is pivotal in enabling the human-centric framework of Industry 5.0. It facilitates adaptive systems, enhances decision-making, and supports predictive maintenance, allowing workers to focus on more complex tasks while AI handles repetitive jobs (Ivanov, 2022). Collaborative robots, for instance, work alongside humans to improve precision and safety, boosting productivity without compromising autonomy or creativity (Boschetti, et al., 2023). However, integrating AI and robotics into the workforce presents challenges, particularly regarding job displacement. To mitigate these risks, continuous reskilling and upskilling of workers must be prioritized (Grabowska et al., 2022). Kolade and Owoseni (2022) advocate for a focus on workforce adaptation, tackling job polarization and addressing inequalities exacerbated by technological disruption. These insights are vital in the context of students' preparedness for future roles in this evolving workforce, underscoring the importance of proactive educational frameworks.

Ethical concerns are central to designing human-centric AI systems. Ensuring algorithmic fairness, transparency, and accountability in AI decision-making is crucial to prevent biases and guarantee equitable outcomes (Binns, 2018). While AI enhances decision-making, operational efficiency, and predictive maintenance (Chowdhury et al., 2023; Elish & Boyd, 2017), it also raises issues related to data privacy, algorithmic transparency, and potential biases (European Commission: Directorate-General for Research and Innovation, 2024). Adel (2022) highlights these ethical concerns as particularly relevant in Industry 5.0, where technological progress must align with human values. Explainable AI (XAI) is a key strategy for addressing these issues, offering transparency and interpretability in AI-driven decisions, which fosters trust and scrutiny (Maddikunta et al., 2022). Integrating Explainable AI (XAI) into Industry 5.0 systems is crucial for fostering worker trust and enabling smooth collaboration between humans and advanced technologies (Toth et al., 2023). The development of comprehensive policy frameworks is also critical to ensure the ethical implementation of AI, focusing on issues such as data privacy, fairness, and aligning AI technologies with sustainability and inclusivity goals (Pacheco & Iwaszczenko, 2024).

Education 5.0 prepares the workforce for Industry 5.0 by integrating AI and IoT, providing students with both technical expertise and essential interpersonal skills for an AI-driven future (Tavares et al., 2022). Education 5.0 focuses not only on technical proficiency but also on fostering critical thinking, creativity, and problem-solving skills, ensuring students are prepared to collaborate with AI in the future workforce. Furthermore, it helps mitigate concerns about job displacement by preparing students to work alongside AI systems, thus shaping a responsible technological future.

Workforce empowerment is a core tenet of Industry 5.0 (Rame et al., 2024). The literature stresses that AI adoption must enhance, not undermine, the value of human workers. AI should empower workers by enabling them to perform higher-value tasks that require creativity, problem-solving, and critical thinking (Alves et al., 2023). AI's role as a human empowerment tool ensures job satisfaction and well-being (Gamberini & Pluchino, 2024). However, the integration of AI must be approached with care to address its ethical, societal, and economic implications. The successful integration of AI requires collaborative efforts among governments, businesses, and workers (Rame et al., 2024). Studies by

Rajumesh (2024) and Adel (2022) reinforce that balancing AI-driven innovation with human empowerment is essential for realizing Industry 5.0's full potential in an equitable and sustainable manner.

This study is theoretically anchored in the Technology Readiness and Acceptance Model (TRAM) (Lin et al., 2007), which integrates individual readiness traits—optimism, innovativeness, discomfort, and insecurity—with perceptions of usefulness and ease of use. This integration enhances the model's predictive capacity for understanding the adoption of emerging intelligent technologies. In this research, TRAM is applied to examine how students' AI literacy, digital self-efficacy, and openness to innovation shape their attitudes toward AI-driven workforce transformation.

Complementing this framework, Trust in Automation Theory (Lee & See, 2004) conceptualizes trust as a function of perceived system reliability, transparency, and ethical alignment—factors critical to user acceptance of automated systems. This theory is particularly suited for analysing students' concerns regarding explainability, algorithmic bias, and decision-making accountability in AI, all of which are central to ethical human-AI collaboration in Industry 5.0.

Research methodology

This study employs a quantitative research design within the positivist paradigm, focusing on university students' perceptions of AI and its potential to shape the future workforce in the context of Industry 5.0.

Sample Characteristics and Data Collection

A total of 344 students participated in the study, drawn from both public and private higher education institutions in Albania. The sample comprised 231 Bachelor's students and 113 Master's students, ensuring representation across academic levels. Disciplinary distribution included 40% Information Technology, 35% Computer Engineering, and 25% Business Management. These students were selected for their potential future roles as technologists, business leaders, and policymakers who will influence the adoption and integration of AI technologies in the evolving workforce. Participants were predominantly aged 20–24 years, with a gender distribution of 36% female and 64% male.

Data collection was carried out through an online survey, ensuring broad accessibility and maintaining the anonymity and confidentiality of the participants. The study complied with institutional research ethics regulations. As it involved an anonymous, voluntary survey without the collection of sensitive or personally identifiable data, formal ethics committee approval was not required.

Survey Instrument Structure

The survey instrument was developed based on an extensive literature review encompassing AI adoption, ethical implications, and Industry 5.0 (Adel, 2022; Alves, et al., 2023; Binns, 2018; Gamberini & Pluchino, 2024; Pacheco & Iwaszczenko, 2024). It was structured into four thematic dimensions, each reflecting key constructs derived from the theoretical framework and aligned with the study's research objectives:

- Demographic and Academic Profile (5 items): This section gathered background information on students' age, gender, level of study (Bachelor or Master), academic discipline (Information Technology, Computer Engineering, or Business), and prior exposure to AI-related topics. These variables provided essential contextual insight for interpreting variation in students' perceptions of AI. The design of this dimension is informed by Adel (2022), who emphasizes educational personalization and the alignment of learning environments with the evolving demands of Industry 5.0.
- Knowledge and Exposure to AI Technologies (8 items): This dimension assessed students' familiarity with core AI technologies, including machine learning, robotics, and natural language processing—designed to measure baseline AI literacy. The structure of this dimension is conceptually grounded in Adel (2022) and Alves et al. (2023), who highlight the central role of digital fluency and AI competence in preparing students for workforce readiness and facilitating effective human–technology integration within the context of Industry 5.0.

- Perceptions of AI in Workforce Transformation (10 items): This dimension assessed students' beliefs about AI's capacity to enhance productivity, drive innovation, and facilitate workforce reskilling. It captured perceptions of AI as an enabler of human performance rather than a replacement. The structure is grounded in the human-centric paradigm of Industry 5.0, as articulated by Gamberini & Pluchino (2024) and Pacheco & Iwaszczenko (2024), who emphasize the role of AI in augmenting human creativity, adaptability, and collaborative potential.
- Ethical Concerns and Transparency Expectations (9 items): This dimension explored students' concerns regarding AI-related risks, including job displacement, algorithmic bias, data privacy, and the need for transparent, explainable systems. It reflects students' ethical awareness and expectations for responsible AI deployment. The theoretical foundation is drawn from Binns (2018), who underscores the importance of algorithmic transparency and fairness, along with Adel (2022) and Pacheco & Iwaszczenko (2024), who advocate for integrating ethical governance into the Industry 5.0 framework.

Statistical Analysis

The reliability of the questionnaire was confirmed using Cronbach's alpha, which yielded a value of 0.94, indicating high internal consistency (Table 1). Data were analysed using JASP (version 0.19.1.0). Descriptive statistics were employed to summarize key patterns across the four dimensions of the survey, offering a broad overview of students' perceptions of AI's role within the context of Industry 5.0. Spearman's rank-order correlation was used to explore the relationships between variables, given the ordinal nature of Likert-scale responses and the non-parametric assumptions required for such analysis (Lovie, 1995). All statistical analyses were conducted at a significance level of $\alpha = 0.05$, with a 95% confidence interval.

This study offers important insights into students' perceptions of AI and its role in shaping the future workforce. While based on a cross-sectional sample from Albanian universities, the inclusion of diverse academic disciplines enhances the study's internal validity. These findings establish a solid foundation for future research to build upon through longitudinal designs and internationally diverse samples, enabling broader generalization and cross-cultural comparison.

Table 1. Cronbach's α (Source: Created by the authors)

Estimate	Cronbach's α
Point estimate	0.945

Research results

How do students perceive AI's potential to influence the future workforce and drive transformation across industries in the context of Industry 5.0?

The analysis of students' perceptions of AI's potential to shape the future workforce and transform industries within the context of Industry 5.0 reveals a mix of cautious optimism and scepticism, with important implications for both educational and industrial strategies (Table 2). The dimension AI's Potential to Transform Industries and Fields recorded the highest overall mean ($M = 3.494$), with Information Technology students ($M = 3.935$) and Computer Engineering students ($M = 3.958$) expressing high confidence, in contrast to Business Management students ($M = 2.14$), whose low score suggests limited awareness or scepticism.

The dimension AI's Role in Enhancing Human Abilities (e.g., creativity, productivity, learning) recorded a mean of $M = 2.875$, with Computer Engineering students scoring highest ($M = 3.075$), followed by Information Technology students ($M = 2.971$), and Business Management students ($M = 2.442$).

The dimension Importance of AI for the Future of Technology and Society recorded a total mean of M = 3.419, with relatively high scores from Information Technology students (M = 3.623) and Computer Engineering students (M = 3.708), and a notably lower mean from Business Management students (M = 2.686).

The dimension Awareness of Industry 5.0 showed the lowest overall mean (M = 2.238), with Computer Engineering students reporting M = 2.392, Information Technology students M = 2.239, and Business Management students M = 2.023.

The dimension Integration of Human Resources and AI in Industry 5.0 resulted in a moderate mean (M = 2.890), with Information Technology students reporting M = 3.087, Computer Engineering students M = 2.975, and Business Management students M = 2.453.

The dimension Confidence in Industry 5.0's Ability to Boost Productivity and Efficiency yielded a total mean of M = 2.878, with Computer Engineering students scoring M = 3.075, Information Technology students M = 3.014, and Business Management students M = 2.384.

The dimension Importance of Workforce Reskilling for Industry 5.0 showed an overall mean of M = 3.186, with Information Technology students scoring M = 3.348, Computer Engineering students M = 3.250, and Business Management students M = 2.837.

The dimension Role of Government and Policy in Implementing Industry 5.0 recorded a total mean of M = 3.137, with Information Technology students scoring M = 3.297, Computer Engineering students M = 3.225, and Business Management students M = 2.756.

Table 2. AI and Industry 5.0. Descriptive Statistics (Source: Created by the authors)

Statement	Valid	Mean				Min	Max
		Information Technology (138)	Computer Engineering (120)	Business Management (86)	Total		
AI's Potential to Transform Industries and Fields	344	3.935	3.958	2.14	3.494	1	5
AI's Role in Enhancing Human Abilities (Creativity, Productivity, Learning)	344	2.971	3.075	2.442	2.875	1	5
Importance of AI for the Future of Technology and Society	344	3.623	3.708	2.686	3.419	1	5
Awareness of Industry 5.0	344	2.239	2.392	2.023	2.238	1	5
Integration of Human Resources and AI in Industry 5.0	344	3.087	2.975	2.453	2.89	1	5
Confidence in Industry 5.0's Ability to Boost Productivity and Efficiency	344	3.014	3.075	2.384	2.878	1	5
Importance of Workforce Reskilling for Industry 5.0	344	3.348	3.25	2.837	3.186	1	5
Role of Government and Policy in Implementing Industry 5.0	344	3.297	3.225	2.756	3.137	1	5

What are students' views on the ethical challenges associated with AI, such as job displacement, privacy risks, and algorithmic bias, within Industry 5.0?

Students' perceptions of the ethical challenges posed by AI in Industry 5.0 reveal significant concerns that reflect the evolving landscape of technology (Table 3). The dimension Transparency and Explainability recorded a mean score of M = 3.055, with concern highest among Computer Engineering students (M = 3.292), followed by Information Technology (M = 3.188), and lowest among Business Management students (M = 2.512).

The dimensions Risks, Manipulation, and Irreversible Consequences recorded the highest overall mean (M = 3.177), with concern most pronounced among Information Technology students (M = 3.370), followed by Computer Engineering (M = 3.292) and Business Management students (M = 2.709).

The dimension Potential Negative Consequences of Industry 5.0 yielded a moderate overall concern (M = 2.962). Information Technology (M = 3.087) and Computer Engineering students (M = 3.083) reported comparable levels of concern, whereas Business Management students expressed lower concern (M = 2.593).

The dimension of Ethical Implications of AI Implementation recorded the lowest overall mean (M = 2.916). Concern was highest among Information Technology students (M = 3.138), followed by Computer Engineering (M = 2.967), while Business Management students reported the lowest level of concern (M = 2.488).

Table 3: Ethical challenges. Descriptive Statistics (Source: Created by the authors)

Statement	Valid	Mean				Min	Max
		Information Technology (138)	Computer Engineering (120)	Business Management (86)	Total		
Transparency and Explainability in AI Systems	344	3.188	3.292	2.512	3.055	1	5
Risks, Manipulation, and Irreversible Consequences	344	3.37	3.292	2.709	3.177	1	5
Potential Negative Consequences of Industry 5.0	344	3.087	3.083	2.593	2.962	1	5
Ethical Implications of Industry 5.0	344	3.138	2.967	2.488	2.916	1	5

H1: Students with greater knowledge of AI technologies will report more positive perceptions of AI's potential to transform industries and enhance human skills within the context of Industry 5.0 ($\alpha=0.05$).

The Spearman's correlation analysis provides empirical support for H1. A statistically significant moderate positive correlation ($\rho = 0.373$, $p < .001$) was observed between students' knowledge of AI technologies (independent variable) and their perceptions of AI's potential to transform industries (dependent variable), indicating that students with higher AI literacy are more likely to view AI as a driver of industrial transformation and productivity enhancement in Industry 5.0.

Moreover, a significant positive correlation ($\rho = 0.289$, $p < .001$) was found between knowledge of AI technologies (independent variable) and students' perceptions of AI's role in enhancing human skills (dependent variable), such as creativity and productivity. These findings suggest that students who are more knowledgeable about AI tend to perceive it not as a threat but as a complement to human capabilities, reinforcing the vision of Industry 5.0, which emphasizes the integration of human and artificial intelligence (Grabowska, et al., 2022; Tavares, et al., 2022).

In conclusion, the results demonstrate that greater AI knowledge is significantly associated with more optimistic perceptions of AI's transformative and augmentative role in future industries. These outcomes underline the value of embedding AI literacy in higher education curricula to better prepare students for technology-integrated, human-centred industrial futures.

Table 4. Spearman's Correlations (Source: Created by the authors)

Variable		Knowledge of AI Technologies	Perception of AI's Potential to Transform Industries	Perception of AI's Role in Enhancing Human Skills	Transparency and Explainability in AI Systems	Concerns About Ethical Implications of AI	Awareness of Industry 5.0	Confidence in Industry 5.0's Impact on Productivity and Efficiency	Integration of Human Resources and AI in Industry 5.0	Importance of Workforce Reskilling for Industry 5.0	Role of Government and Policy in Implementing Industry 5.0	Perceived Risks, Manipulation, Irreversible Consequences	Perceived Negative Consequences of Industry 5.0
Knowledge of AI Technologies	n	—											
	Spearman's rho	—											
	p-value	—											
Perception of AI's Potential to Transform Industries	n	344	—										
	Spearman's rho	0.373***	—										
	p-value	< .001	—										
Perception of AI's Role in Enhancing Human Skills	n	344	344	—									
	Spearman's rho	0.289***	0.272***	—									
	p-value	< .001	< .001	—									
Transparency and Explainability in AI Systems	n	344	344	344	—								
	Spearman's rho	0.299***	0.492***	0.414***	—								
	p-value	< .001	< .001	< .001	—								
Concerns About Ethical Implications of AI	n	344	344	344	344	—							
	Spearman's rho	0.359***	0.347***	0.358***	0.295***	—							
	p-value	< .001	< .001	< .001	< .001	—							
	n	344	344	344	344	344	—						
Awareness of Industry 5.0	Spearman's rho	0.199***	0.135*	0.143**	0.092*	0.258***	—						
	p-value	< .001	0.012	0.008	0.039	< .001	—						

Confidence in Industry 5.0's Impact on Productivity and Efficiency	N	344	344	344	344	344	344	—					
	Spearman's rho	0.307***	0.392***	0.219***	0.311***	0.529***	0.298***	—					
	p-value	< .001	< .001	< .001	< .001	< .001	< .001	—					
Integration of Human Resources and AI in Industry 5.0	n	344	344	344	344	344	344	344	—				
	Spearman's rho	0.305***	0.359***	0.201***	0.227***	0.417***	0.306***	0.675***	—				
	p-value	< .001	< .001	< .001	< .001	< .001	< .001	< .001	—				
Importance of Workforce Reskilling for Industry 5.0	n	344	344	344	344	344	344	344	344	—			
	Spearman's rho	0.285***	0.347***	0.214***	0.362***	0.545***	0.287***	0.599***	0.58***	—			
	p-value	< .001	< .001	< .001	< .001	< .001	< .001	< .001	< .001	—			
Role of Government and Policy in Implementing Industry 5.0	n	344	344	344	344	344	344	344	344	344	—		
	Spearman's rho	0.264***	0.358***	0.165**	0.314***	0.481***	0.185***	0.494***	0.579***	0.668***	—		
	p-value	< .001	< .001	0.002	< .001	< .001	< .001	< .001	< .001	< .001	—		
Perceived Risks, Manipulation, Irreversible Consequences	n	344	344	344	344	344	344	344	344	344	344	—	
	Spearman's rho	0.345***	0.434***	0.168**	0.413***	0.374***	0.092*	0.309***	0.214***	0.391***	0.326***	—	
	p-value	< .001	< .001	0.002	< .001	< .001	0.037	< .001	< .001	< .001	< .001	—	
Perceived Negative Consequences of Industry 5.0	n	344	344	344	344	344	344	344	344	344	344	344	—
	Spearman's rho	-0.21***	0.322***	-0.113*	0.295***	0.659***	0.168***	0.443***	0.411***	0.524***	0.564***	0.348***	—
	p-value	< .001	< .001	0.036	< .001	< .001	0.002	< .001	< .001	< .001	< .001	< .001	—

* p < .05, ** p < .01, *** p < .001

H2: Students with stronger concerns about the ethical implications of AI are more likely to view transparency and explainability in AI systems as critical to the success of Industry 5.0 ($\alpha = 0.05$).

The Spearman's correlation analysis provides strong empirical support for H2. A statistically significant positive correlation ($\rho = 0.295$, $p < .001$) was observed between concerns about the ethical implications of AI (independent variable) and the perceived importance of transparency and explainability in AI systems (dependent variable) (Table 4). This finding indicates that students who express greater concern about ethical risks—such as algorithmic bias, privacy violations, and job displacement—are more likely to emphasize the need for transparent and interpretable AI systems in the context of Industry 5.0.

This aligns with existing literature that emphasizes transparency as a foundational element in fostering trust, fairness, and accountability in AI applications (Binns, 2018). In human-centric frameworks like Industry 5.0, where AI technologies are deeply embedded in critical sectors, transparent AI design becomes essential for mitigating misuse and reinforcing ethical standards (Grabowska, et al., 2022; Toth, et al., 2023).

These findings underscore the importance of integrating transparency and explainability as core design principles in AI development. As AI continues to transform industries under the paradigm of Industry 5.0, addressing ethical concerns through clear, interpretable, and trustworthy systems will be vital for ensuring public acceptance and responsible innovation. Students' prioritization of transparency reflects a broader societal demand for ethical and human-centred technological governance.

This study offers an empirically grounded exploration of university students' perceptions of AI and human-centric technologies in shaping the future workforce within the Industry 5.0 paradigm. By applying the Technology Readiness and Acceptance Model (TRAM) (Lin et al., 2007) and Trust in Automation Theory (Lee & See, 2004), the research reveals how students' behavioural traits, cognitive trust mechanisms, and ethical orientations intersect to inform their attitudes toward AI integration.

The results confirm that AI knowledge significantly correlates with positive perceptions of AI's transformative role ($\rho = 0.373$, $p < .001$), reinforcing the TRAM hypothesis that optimism and digital fluency enhance acceptance of intelligent systems. This is particularly evident among Information Technology and Computer Engineering students, who scored markedly higher across all dimensions compared to their Business Management peers. Such disciplinary disparities suggest unequal exposure to AI curricula, which may influence both confidence in AI and readiness for future roles. These findings align with Grabowska et al. (2022) and Gamberini & Pluchino (2024), who emphasize the importance of integrating technical training with innovation readiness in shaping workforce preparedness.

In addition, Trust in Automation Theory is validated through the strong correlation between ethical concerns and the prioritization of AI transparency and explainability ($\rho = 0.295$, $p < .001$). Students who are more attuned to the risks of AI—such as bias, manipulation, and data misuse—tend to demand higher accountability in AI design. This supports prior research (Binns, 2018; Toth et al., 2023), which identifies transparency as a cornerstone of trust and responsible deployment in human-AI collaboration.

Another important finding is the low overall awareness of Industry 5.0, particularly among Business Management students ($M = 2.023$). This points to a critical knowledge gap in non-technical disciplines and underscores the need for interdisciplinary education that incorporates ethical, societal, and technological perspectives. As Education 5.0 advocates (Tavares et al., 2022) argue, preparing the future workforce requires bridging technical and humanistic domains to foster adaptability and critical thinking alongside digital skills.

Collectively, these findings emphasize that students' perceptions are not solely shaped by their knowledge of AI, but also by their trust in its ethical governance and societal alignment. As such, this research extends the application of TRAM and Trust in Automation Theory beyond traditional industrial settings, demonstrating their relevance in academic and socio-cultural contexts within developing economies.

Conclusions

This study contributes to the growing discourse on Industry 5.0 by highlighting how university students - future technologists, policymakers, and business leaders - perceive the integration of AI into the evolving workforce. The evidence reveals that students who possess greater AI literacy exhibit stronger optimism about AI's potential to drive innovation, enhance human productivity, and enable new industrial roles. However, the findings also underscore substantial concerns related to ethical implications, particularly algorithmic fairness, job displacement, and transparency.

Crucially, this research affirms that technical knowledge alone is not sufficient; ethical considerations and trust are equally central to students' acceptance of AI. The emphasis placed on transparency and explainability suggests that ethical governance frameworks must be embedded not only in AI systems but also in education strategies. These insights reinforce the necessity for Education 5.0 frameworks that blend technical proficiency with socio-ethical competencies, ensuring that students are prepared to engage with AI in both critical and constructive ways.

From a policy and curriculum development perspective, the results advocate for inclusive and cross-disciplinary AI education that empowers all students—regardless of academic background—to navigate the challenges and opportunities of human-AI collaboration. Addressing gaps in awareness, especially among non-technical disciplines, will be vital for fostering equitable and widespread workforce readiness.

In sum, as the principles of Industry 5.0 continue to shape the contours of future work, the formation of an ethically aware, technically proficient, and human-centred workforce will depend not only on innovation, but on education that cultivates trust, transparency, and adaptability. This study provides a regional contribution to this global challenge, offering actionable insights for stakeholders in education, governance, and industry.

References

Adel, A. (2022). Future of Industry 5.0 in society: Human-centric solutions, challenges and prospective research areas. *Journal of Cloud Computing*, *11*(1), 40. https://doi.org/10.1186/s13677-022-00314-5

Alves, J., Lima, T. M., & Gaspar, P. D. (2023). Is Industry 5.0 a human-centred approach? A systematic review. *Processes*, *11*(1), 193. https://doi.org/10.3390/pr11010193

Binns, R. (2018). Algorithmic accountability and public reason. *Philosophy & Technology*, *31*, 543-556. https://doi.org/10.1007/s13347-017-0263-5

Boschetti, G., Faccio, M., & Granata, I. (2023). Human-centered design for productivity and safety in collaborative robots cells: A new methodological approach. *Electronics*, *12*(1), 167. https://doi.org/10.3390/electronics12010167

Chowdhury, S., Dey, P., Joel-Edgar, S., Bhattacharya, S., Rodriguez-Espindola, O., Abadie, A., & Truong, L. (2023). Unlocking the value of artificial intelligence in human resource management through AI capability framework. *Human Resource Management Review*, *33*(1), 100899. https://doi.org/10.1016/j.hrmr.2022.100899

Elish, M. C., & Boyd, D. (2017). Situating methods in the magic of Big Data and AI. *Communication Monographs*, *85*(1), 57-80. https://doi.org/10.1080/03637751.2017.1375130

European Commission (n.d.). *What is Industry 5.0?* https://research-and-innovation.ec.europa.eu/research-area/industrial-research-and-innovation/industry-50_en

European Commission: Directorate-General for Research and Innovation (2024). *ERA industrial technologies roadmap on human-centric research and Innovation for the manufacturing sector.* Publications Office of the European Union. https://data.europa.eu/doi/10.2777/0266

Gamberini, L., & Pluchino, P. (2024). Industry 5.0: A comprehensive insight into the future of work, social sustainability, sustainable development, and career. *Australian Journal of Career Development*, *33*(1), 5-14. https://doi.org/10.1177/10384162241231118

Ghobakhloo, M., Iranmanesh, M., Fathi, M., Rejeb, A., Foroughi, B., & Nikbin, D. (2024). Beyond Industry 4.0: A systematic review of Industry 5.0 technologies and implications for social, environmental and economic sustainability. *Asia-Pacific Journal of Business Administration*, ahead-of-print. https://doi.org/10.1108/APJBA-08-2023-0384

Giroux, M., Kim, J., Lee, J. C., & Park, J. (2022). Artificial intelligence and declined guilt: Retailing morality comparison between human and AI. *Journal of Business Ethics*, *178*, 1027-1041. https://doi.org/10.1007/s10551-022-05056-7

Grabowska, S., Saniuk, S., & Gajdzik, B. (2022). Industry 5.0: Improving humanization and sustainability of Industry 4.0. *Scientometrics*, *127*, 3117-3144. https://doi.org/10.1007/s11192-022-04370-1

Ivanov, D. (2022). The Industry 5.0 framework: Viability-based integration of the resilience, sustainability, and human-centricity perspectives. *International Journal of Production Research*, *61*(5), 1683-1695. https://doi.org/10.1080/00207543.2022.2118892

Kolade, O., & Owoseni, A. (2022). Employment 5.0: The work of the future and the future of work. *Technology in Society*, *71*, 102086. https://doi.org/10.1016/j.techsoc.2022.102086

Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *46*(1), 50-80. https://doi.org/10.1518/hfes.46.1.50_30392

Leng, J., Sha, W., Wang, B., Zheng, P., Zhuang, C., Liu, Q., Wuest, T., Mourtzis, D., & Wang, L. (2022). Industry 5.0: Prospect and retrospect. *Journal of Manufacturing Systems*, *65*, 279-295. https://doi.org/10.1016/j.jmsy.2022.09.017

Lin, C.-H., Shih, H.-Y., & Sher, P. J. (2007). Integrating technology readiness into technology acceptance: The TRAM model. *Psychology & Marketing*, *24*(7), 641-657. https://doi.org/10.1002/mar.20177

Liu, S., Li, P., Wang, J., & Liu, P. (2024). Toward Industry 5.0: Challenges and enablers of intelligent manufacturing technology implementation under the perspective of sustainability. *Heliyon*, *10*(15), e35162. https://doi.org/10.1016/j.heliyon.2024.e35162

Longo, F., Padovano, A., & Umbrello, S. (2020). Value-oriented and ethical technology engineering in Industry 5.0: A human-centric perspective for the design of the factory of the future. *Applied Sciences*, *10*(12), 4182. https://doi.org/10.3390/app10124182

Lovie, A. D. (1995). Who discovered Spearman's rank correlation? *British Journal of Mathematical and Statistical Psychology*, *48*(2), 255–269. https://doi.org/10.1111/j.2044-8317.1995.tb01063.x

Lu, Y., Zheng, H., Chand, S., Xia, W., Liu, Z., Xu, X., Wang, L., Qin, Z., & Bao, J. (2022). Outlook on human-centric manufacturing towards Industry 5.0. *Journal of Manufacturing Systems*, *62*, 612-627. https://doi.org/10.1016/j.jmsy.2022.02.001

Maddikunta, P. K. R., Pham, Q.-V., B., Prabadevi, Deepa, N., Dev, K., Gadekallu, T. R., Ruby, R., & Liyanage, M. (2022). Industry 5.0: A survey on enabling technologies and potential applications. *Journal of Industrial Information Integration*, *26*, 100257. https://doi.org/10.1016/j.jii.2021.100257

Pacheco, D. A. de J., & Iwaszczenko, B. (2024). Unravelling human-centric tensions towards Industry 5.0: Literature review, resolution strategies and research agenda. *Digital Business*, *4*(2), 100090. https://doi.org/10.1016/j.digbus.2024.100090

Pizon, J., & Gola, A. (2023). Human - machine relationship - perspective and future roadmap for Industry 5.0 Solutions. *Machines*, *11*(2), 203. https://doi.org/10.3390/machines11020203

Rajumesh, S. (2024). Promoting sustainable and human-centric Industry 5.0: A thematic analysis of emerging research topics and opportunities. *Journal of Business and Socio-economic Development*, *4*(2), 111-126. https://doi.org/10.1108/JBSED-10-2022-0116

Rame, R., Purwanto, P., & Sudarno, S. (2024). Industry 5.0 and sustainability: An overview of emerging trends and challenges for a green future. *Innovation and Green Development*, *3*(4), 100173. https://doi.org/10.1016/j.igd.2024.100173

Tavares, M. C., Azevedo, G., & Marques, R. P. (2022). The challenges and opportunities of Era 5.0 for a more humanistic and sustainable society - a literature Review. *Societies*, *12*(6), 149. https://doi.org/10.3390/soc12060149

Toth, A., Nagy, L., Kennedy, R., Bohus, B., Abonyi, J., & Ruppert, T. (2023). The human-centric Industry 5.0 collaboration architecture. *MethodsX*, *11*, 102260. https://doi.org/10.1016/j.mex.2023.102260